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Rotman

Algorithmic Market Making Case

Rotman International Trading Competition 2026

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OVERVIEW

The Algorithmic Market Making Case is designed to challenge participants' programming skills by developing algorithms using the RIT API to automate trading strategies and react to changing market conditions driven by news releases. Throughout the case, these algorithms will need to provide liquidity while managing risk from sudden price movements. Participants must balance profit-seeking market making activities with strict position limits. Due to the high-frequency nature of the case, participants are encouraged to develop algorithms that can adapt to rapid changes in market dynamics using their selected programming languages.

DESCRIPTION

There will be 12 heats with 1 team member competing in each heat. Only one team member shall trade to represent the team for all heats. Each heat will be 5 minutes long and represent one trading week (5 trading days), with each minute representing one full trading day.

Parameter	Value
Number of trading heats	12
Trading time per heat	5 minutes (300 seconds)
Calendar time per heat	One trading week (5 days)

Order submission using the RIT API will be enabled. Data retrieval via Real-time Data (RTD) Links and the RIT API will also be enabled. **All trades must be executed by a trading algorithm**. Participants will not be allowed to trade manually through the RIT User Application once the heat begins. Participants are allowed to modify their algorithms in response to prevailing market conditions and competition from the algorithms of other teams. They will have 2 minutes between each heat to re-load their algorithms. A base template algorithm will be provided to participants and can be directly modified for use in the competition. However, participants are strongly encouraged to create their own algorithms.

MARKET DYNAMICS

This case involves four stocks with the following details.

Ticker	SPNG	SMMR	ATMN	WNTR
Security type	Stock	Stock	Stock	Stock
Quote currency	CAD	CAD	CAD	CAD
Starting Price	$25	$25	$25	$25
Fee/share (Market orders)	$0.02	$0.02	$0.02	$0.02

Rebate/share (Limit/Passive orders)	$0.01	$0.02	$0.015	$0.025
Max order size	10,000	10,000	10,000	10,000

Your desk is market-making in four stocks, and you expect a volatile week. After each market close (i.e., every minute), your team anticipates a potential news release that may affect these stocks. While the news will not be provided to you directly, your analyst team has indicated that these releases will be the primary driver of price movements throughout the week.

The magnitude and direction of each stock's reaction to a given news event may differ from minute to minute. Some stocks may react more strongly than others, and correlations among stock movements may vary over time.

Given this potential volatility, the Chief Risk Officer (CRO) has imposed an aggregate position limit to control overnight market risk exposure. This limit will be announced at the beginning of the week (i.e., at the start of each heat) and is calculated based on your aggregate position across all four stocks:

$$\begin{aligned} &Aggregate\ Position\ Limit \\ &\qquad = |Position_{\text{SPNG}}| + |Position_{\text{SMMR}}| + |Position_{\text{ATMN}}| + |Position_{\text{WNTR}}| \end{aligned}$$

Penalty for Exceeding Limit: $10 per share for every share above the limit, assessed immediately at each market close (i.e., every minute).

Example: If your aggregate position limit is 15,000 shares and you hold:

- Long 12,000 shares of SPNG
- Short 3,000 shares of SMMR
- Long 7,000 shares of ATMN
- Long 6,000 shares of WNTR

Your aggregate position = | 12,000 | + | -3,000 | + | 7,000 | + | 6,000 | = 28,000 shares
You are 13,000 shares over the limit, incurring a penalty of 13,000 × $10 = $130,000

This penalty structure encourages aggressive market making (providing liquidity on both sides) while penalizing excessive directional risk-taking and suboptimal position management.

TRADING/POSITION LIMITS AND TRANSACTION COSTS

In addition to the aggregate position limit for every market close, each trader will be subject to gross and net trading/position limits during trading in each heat. The gross limit reflects the sum of the absolute values of the long and short positions across all securities, and the net limit reflects the sum of long and short positions such that short positions negate any long positions. Anyone who exceeds the trading/position limits will incur a penalty of $5 per share.

The maximum trade size will be 10,000 shares per order for each stock. Transaction fees will be set at $0.02 per share for each security on all market orders filled. A rebate will be provided for all submitted limit orders that are filled.

POSITION CLOSE-OUT

Any non-zero position of stocks will be closed out at the end of trading based on the last traded price for each stock.

KEY OBJECTIVES

Objective 1

Develop an algorithmic market making strategy that continuously quotes bid and ask prices to profit from the bid-ask spread while managing inventory risk. Your algorithm should provide liquidity to the market while avoiding excessive position accumulation.

Objective 2

Manage total exposure across all four stocks, potentially by flattening positions as they approach the Limit, balancing long and short positions across different stocks, and implementing risk controls that prevent excessive position accumulation.

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Scoring Methodology

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OVERVIEW

The scoring and ranking methodology is designed to translate absolute performance into relative performance by the use of a ranking system. This ranking system is designed to discourage participants from "betting the house" in one sub-heat and generating very large absolute profits that will result in a clear win of the entire competition. Instead, participants' absolute performance in each sub-heat is converted into a series of ordinal ranks which are subsequently converted into a final case ranking. These case rankings are mapped to case scores and then combined under the following weights:

Case	Weight
Liquidity Risk Case	20%
Volatility Trading Case	20%
GBE Energy Electricity Trading Case	20%
Merger Arbitrage Case	20%
Algorithmic Marketing Making Case	20%

The scoring system is not intended to be extremely complex. However, throughout the trading competition there will be over 2,000 separate trading results. These results must then be averaged and ranked over several iterations to compute a final ranking and score. This document describes that process.

The purpose of the system is to reward consistently high performance (i.e. a team that places 8th, 5th, and 10th will have a higher final score than a team that places 1st, 10th, and 35th).

Liquidity Risk Case, Volatility Trading Case and Merger Arbitrage Case

For each sub-heat, the final profits and losses (P&L)[5] of all participating members of a team are summed to form a dollar value of the team P&L. The teams are then ranked for each sub-heat by the dollar values of the team P&L with 1st place given to the team with the highest dollar value. In the event of a tie, the teams that have tied will be given the same rank. The teams below the tie will be given a rank based on the number of teams that have scored better than them. Therefore, if three teams tied for 2nd place, the ranking for the top five teams would be 1st, 2nd, 2nd, 2nd, and 5th.

Each team's sub-heat ranks are then averaged. Teams are then ranked based on their average sub-heat rank to determine their final case rank. The team with the lowest average will be ranked first.

This case ranking is then mapped to a point score where the lowest rank (best score) is given a score of n+1, where n is the number of teams below you plus the teams that tied with you (i.e. the first place team out of 39 teams will get a score of 39, the last place team will get a score of 1). To continue the above example, if you are tied for 2nd place with three other teams, you will get a score of 38.

[5] Adjusted for the relevant case-specific penalties as described in each case.

GBE Energy Electricity Trading Case

The final P&L of each team member will be summed to form a dollar value of the team P&L. The teams are then ranked for each sub-heat by the dollar values of the team P&L, with first place awarded to the team with the highest dollar value. In the event of a tie, the teams that have tied will be given the same rank. The teams below the tie will be given a rank based on the number of teams that have scored better than them. Therefore, if three teams tied for 2nd place, the ranking would be 1st, 2nd, 2nd, 2nd, and 5th.

Based on the above, each team's sub-heat ranks will be averaged and then the resulting averages will be ranked to determine their overall case rank. The team with the lowest average will be ranked first. This case ranking is then mapped to a point score where the lowest rank is given a score of n+1, where n is the number of teams below you plus the teams that tied with you.

Algorithmic Market Making Case

Only one member from each team will be required to participate in the Algorithmic Market Making Case. The final P&L[6] of the participating team member will become the team P&L, which will be then ranked for each heat with first place awarded to the team with the highest dollar value. In the event of a tie, the teams that have tied will be given the same rank. The teams below the tie will be given a rank based on the number of teams that have scored better than them. Therefore, if three teams tied for 2nd place, the ranking would be 1st, 2nd, 2nd, 2nd, and 5th.

Based on the above, each team's heat ranks will be averaged and then the resulting averages will be ranked to determine the final case rank. The team with the lowest average will be ranked first. This case ranking is then mapped to a point score where the lowest rank is given a score of n+1, where n is the number of teams below you plus the teams that tied with you.

Final Score

The final case scores are then multiplied by their case-weights to form a final weighted score. This final weighted score is used to rank teams, where the highest score is the best score. In the case of two or more teams having the same final weighted score, those teams will be ranked based on the variance of their final case scores. The team with the lowest variance will be ranked ahead of the others. For example, if the top 3 teams have the following scores:

[6] Adjusted for the relevant case-specific penalties as described in the case package.

Team	Final Case Scores					Final Weighted Score
	Liquidity Risk Case	Volatility Trading Case	GBE Energy Electricity Trading Case	Merger Arbitrage Case	Algorithmic Marketing Making Case	
Team 1	28	25	27	21	25	25.2
Team 2	27	27	21	22	23	24
Team 3	27	23	25	24	21	24

Team 1 will be ranked first as it has the highest weighted score. Team 2 and Team 3 have the same final weighted score and will be ranked based on the variance of their case scores. The variance for Team 2 is 8 while the variance for Team 3 is 5, therefore Team 3 will be ranked second while Team 2 will be ranked third.

Final Rank	Team
1	Team 1
2	Team 3
3	Team 2

Two (or more) teams that have the same score and the same variance will tie. In the event of a tie, the teams that have tied will be given the same rank. The teams below the tie will be given a rank based on the number of teams that have scored better than them. Therefore, if three teams tied for 2nd place, the ranking would be 1st, 2nd, 2nd, 2nd, and 5th.

Awards

The top five ranking teams will receive the following cash prizes:

- 1st place team: $5,000 CAD
- 2nd place team: $2,500 CAD
- 3rd place team: $1,500 CAD
- 4th place team: $1,000 CAD
- 5th place team: $500 CAD